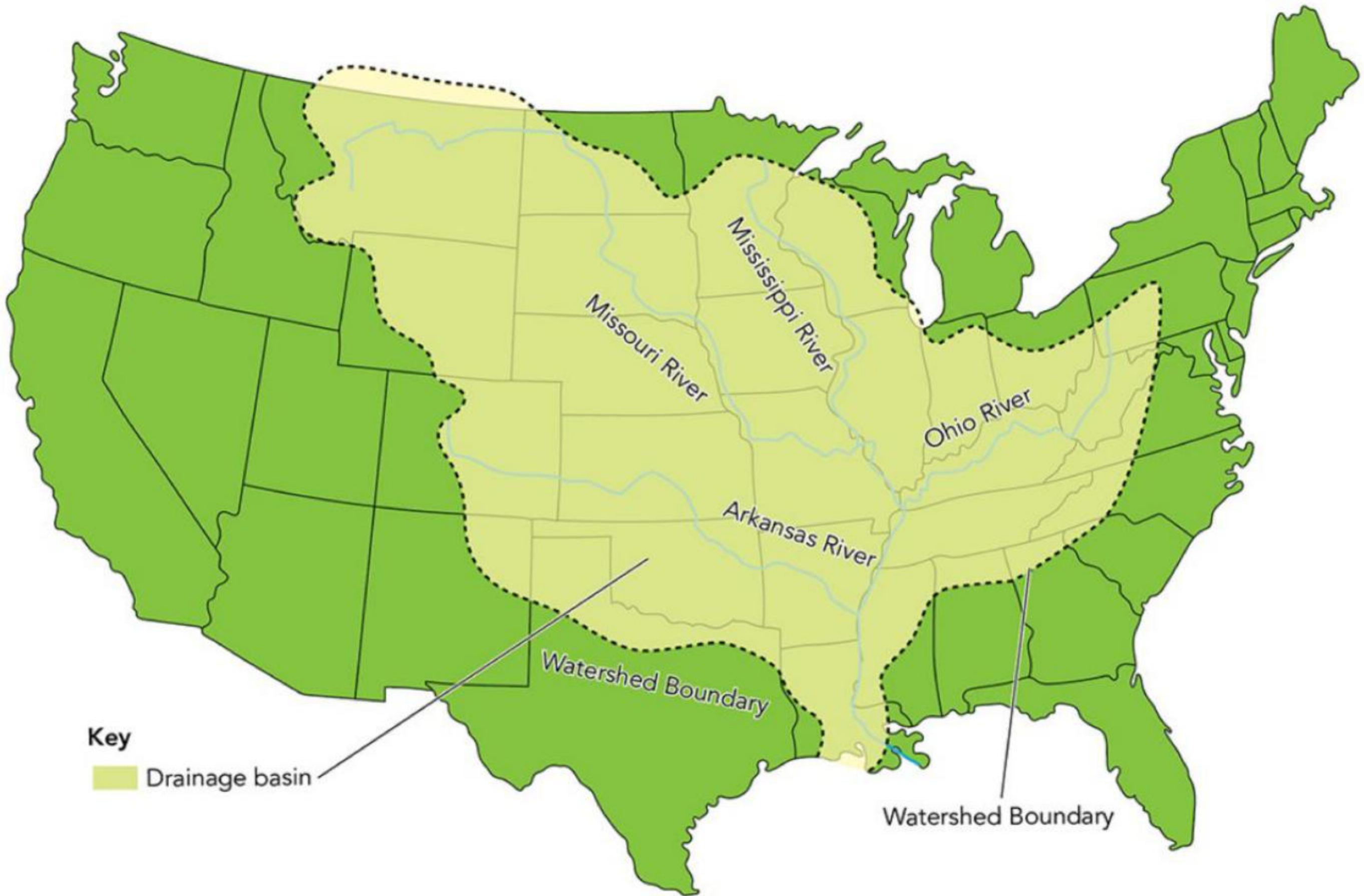


1.5 Detailed Specific Example: The Mississippi River, USA



The Mississippi River in the USA is the largest river in the North American continent. Its drainage basin extends over 31 states, covering around 15% of the land area of North America. The river and surrounding basin are economically important; they are used for agriculture, transportation, and the provision of clean water and energy.



Source **A**: The Mississippi River basin

The Mississippi River experiences frequent and sometimes devastating floods. This is not a new trend; flooding has always taken place along the Mississippi, and the Great Mississippi Flood of 1927 remains one of the most destructive in the country’s history.

TASK 1: Study Source **A**

Discuss in pairs why the Mississippi’s tributaries contribute to flooding downstream.

Source **B**: Important Mississippi flood events over time

Year	Dates/Months	Details
1200		Archaeological records suggest the Native American settlement of Cahokia was devastated as a result of flooding.
1543		Explorers encountered a flood lasting over 40 days.

1734 to 1735	December to June	City of New Orleans was flooded for several months.
1851	June	Major flooding due to excessive rains in Iowa.
1874	February to May	The Great Mississippi Flood of 1874. 375 000 homes and over 12 000 000 acres of land affected.
1882	February	Around 20 000 people made homeless in the state of Arkansas alone.
1912	March to May	Heavy snowfall quickly melted, which caused extensive flooding along the Mississippi and its tributaries.
1913	April to May	One of the worst floods in US history. It is estimated that around 1000 people lost their lives.
1927	January to May	The Great Mississippi Flood of 1927. Floods lasting 135 days. Persistent rainfall caused floodwaters to span nearly 130 km in places.
1937	February to March	The Great Mississippi Flood of 1937. Heavy rainfall and snowmelt led to the Mississippi river exceeding previous flood records.
1973	March to May	The largest volume of water to flow down the Mississippi since the 1927 flood.
2011	March to June	The Great Mississippi Flood of 2011. Caused by snowmelt and heavy rains in the Ohio Valley.
2019	Dec 2018 to Aug 2019	The longest known flood on record of the lower Mississippi River, lasting for 226 days.

TASK 2: Study Source B

- a **Create a bar graph** to show the months that have been affected by flooding over time. For example, the 1874 flood affected February, March, April, and May.
- b Which months are the most likely to be affected by flooding?
- c Which months are the least likely to be affected by flooding?
- d Suggest why some months have a greater potential for flooding than others.

Causes of flooding

Natural causes of flooding along the Mississippi include heavy rainfall and snowmelt, particularly during the spring months. The river has numerous upstream tributaries, including the Arkansas River and the Ohio River. The volume of water flowing along these tributaries adds to rising water levels.

Human causes of flooding include urbanisation and deforestation, which lead to an increase in surface run-off. An estimated 70 million people live within the Mississippi drainage basin in cities such as New Orleans, Minneapolis, and Memphis. Large areas of land have been cleared to make way for agriculture. In addition to this, the impacts of climate change have made the natural causes of flooding worse. (see [Topic 5](#) for more about this).

Impacts of flooding

Flooding along the Mississippi leads to property and infrastructure damage, loss of life, displacement of communities, and loss of income. Flooding can have social, economic, and environmental consequences, for example:

- The Great Mississippi Flood of 1927 caused 246 deaths and left over 700 000 people homeless. The damage to property was estimated to have cost over 400 million.
- During the 2011 flood, deer, wild turkeys, alligators, spiders, rats, and even fire ants were swept away by the waters. The Tennessee government released guidance for residents to avoid displaced snakes.
- The 2019 flood prevented 5.2 million tons of corn and 1.1 million tons of soybeans (valued at \$973 million) from being transported.



Source **C**: Flooded farmland in the Mississippi Basin



Source **D**: Flooded infrastructure in the Mississippi Basin

TASK 3: Study Sources **C** **D**

Explain the social, economic, and environmental costs caused by this flood.

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PROJECT

Work in pairs or small groups to design a flood-resistant infrastructure or housing concept. Research elements like raised foundations, drainage systems, or materials resilient to water damage. Develop a plan and be prepared to present your ideas to the class in a 5-minute pitch. Use sketches to support your presentation and explain the benefits of your proposed solution.

Traditional management techniques

Since the 1700s, attempts to control flooding in the Mississippi Basin have involved the use of levees and floodwalls to increase the capacity and speed of the river. These techniques are controversial because they prevent the river from flowing onto the floodplain, which can push greater volumes of water to areas downstream.

Between 1929 and 1931, the Bonnet Carré Spillway was constructed close to the city of New Orleans. The spillway can divert 250 000 cubic feet of water per second away from the river into nearby Lake Pontchartrain and out to the Gulf of Mexico.

Sustainable management strategies and techniques

In recent years there has been a move towards sustainable approaches to flood management along the Mississippi. Sustainable management strategies and

techniques work with, rather than against, nature. They include:

- Floodplain management: restricting construction in flood-prone areas and restoring natural floodplains and wetlands. A recent study in Illinois found that losing just 1% of wetlands can increase flood volumes by as much as 7%. State and federal governments have bought frequently flooded agricultural areas of floodplain to use as flood storage areas. Since 2019, landowners have been able to also apply for compensation to convert farmland back to floodplains and wetlands.
- Drainage basin management: focusing on land-use practices, such as **reforestation**, which decrease run-off and reduce the volume of water reaching the river. The Wetland Reserve Enhancement Partnership (WREP) programme offers incentives to encourage landowners to carry out reforestation.
- Early warning systems: monitoring and prediction systems can provide flood alerts, enabling people to prepare. The US National Weather Service has webpages providing information about current and potential river levels and flood risk along the Mississippi.
- Community resilience: developing flood-resistant infrastructure and creating disaster preparedness plans. The Federal Emergency Management Agency (FEMA), part of the US government, has a Building Resilient Infrastructure and Communities (BRIC) programme. This supports projects such as the Greenprint for Resilience project in Shelby County, Tennessee, which has created elevated roads and bought vulnerable homes to convert land back to wetlands.

TIP

When learning about sustainable flood management, focus on real-world examples to understand how different strategies and techniques have been successful.

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Figure 1.10: Wetlands store water at times of high flow

‘It has to be dealt with’: what’s the future for midwest towns affected by climate change?

The city of Davenport is grappling with more than 50 days of flooding after the Mississippi spilled over its banks, broke through a barrier and washed into downtown. The streets of smaller cities to the south, such as Buffalo, were underwater for days. “Between hurricanes, tornadoes, floods and droughts, literally all within a year, something’s changing,” said Davenport’s mayor, Frank Klipsch.

Klipsch said the old response would have been to build a higher flood wall. “But you step back a little and that isolationism isn’t necessarily the best approach today,” said Klipsch. “The river is not one city but a whole system in of itself so that whatever one city does it impacts others along the river. One of our small cities, Buffalo, just south of us, had tremendous flood problems now. If we were to put up a flood protection wall it would basically funnel an excessive amount of water and would literally blow that community away.”

- a** Suggest how climate change affects flooding along the Mississippi.
- b** Explain why Davenport’s mayor doesn’t think a higher flood wall is the best approach to deal with the flooding.